

Categorical Time-Series Analysis

0.1 General Setup

0.1.1 Notation

Let $(X_t)_{t \in \mathbb{Z}}$ be a *categorical* time series process, where each X_t takes values in

$$S := \{s_0, \dots, s_m\}, \quad m \in \mathbb{N}.$$

(X_t) may be *nominal*, in which case the ordering of the elements of S is arbitrary and used solely for notational consistency, or *ordinal*, in which case the ordering of the elements of S reflects an inherent order of the categories.

0.1.2 Stationarity

Two forms of stationarity are distinguished. A process is called *weakly stationary* if

$$\begin{aligned} \mathbb{E}(X_t) &= \mu_X \quad \forall t, \\ \mathbb{E}(X_t^2) &< \infty \quad \forall t, \\ \text{Cov}(X_t, X_{t+h}) &= \gamma(h) \quad \forall t, h. \end{aligned}$$

It is called *strictly stationary* if all finite-dimensional distributions are invariant under time shifts, i.e.,

$$F_{X_t, \dots, X_{t+j}} = F_{X_{t+h}, \dots, X_{t+h+j}} \quad \forall t, j, h.$$

0.2 Marginal Distributions

Regardless of whether one considers a simple categorical process with few categories or a complex temporally evolving system, the analysis of categorical processes always rests on the examination of the marginal probabilities. Because ordinal and nominal data encode different types of information, their marginals are represented in different ways.

Ordinal data. Most analytical tools for ordinal processes exploit the natural ordering of the elements of S . For this reason, the marginal distribution is represented by the cumulative distribution function (CDF), which accumulates probabilities in accordance with the ordering of the categories:

$$\mathbf{f} = (f_0, \dots, f_{m-1})^\top, \quad f_i := P(X_t \leq s_i).$$

It suffices to consider the cumulative probabilities of categories $s_0, \dots, s_{m-1}$, since $P(X_t \leq s_m) = 1$ by definition. Although one could equivalently estimate the PMF and derive the CDF from it, the CDF representation is often more convenient because it is directly used to compute location and dispersion measures.

Nominal data. Since nominal categories do not possess a meaningful order, statistical procedures cannot exploit cumulative structure. The marginal distribution is therefore represented by the probability mass function (PMF):

$$\mathbf{p} = (p_0, \dots, p_m)^\top, \quad p_i := P(X_t = s_i).$$

0.3 Population Measures

0.3.1 Location

For a *nominal* process any measure that relies on ordering is inappropriate. The *mode*, defined as the most probable category,

$$\text{Mod}(X_t) = \arg \max_{s_i \in S} P(X_t = s_i),$$

is therefore the natural choice, providing a meaningful summary by identifying the most frequent category.

For an *ordinal* process the natural ordering permits additional measures of central tendency. While the mode remains informative, the *median* respects the ordering and is often more useful for describing the central location:

$$\text{Med}(X_t) = \min \left\{ s_i \in S : F(s_i) \geq \frac{1}{2} \right\}.$$

In summary, the mode identifies the most typical category, whereas the median indicates the midpoint on the ordered scale.

0.3.2 Dispersion

The concept of dispersion differs between ordinal and nominal processes. Two separate indices are used: the *Index of Ordinal Variation* (IOV) for ordinal data and the *Index of Qualitative Variation* (IQV) for nominal data. Both are scaled to $[0, 1]$, where 0 denotes minimal dispersion and 1 denotes maximal dispersion:

$$\text{IOV}(\mathbf{f}) = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i), \quad \text{IQV}(\mathbf{p}) = \frac{m+1}{m} \left(1 - \sum_{i=0}^m p_i^2 \right).$$

Minimal Dispersion

Both measures attain the value 0 if and only if the distribution is degenerate, i.e., all probability mass is concentrated in a single category.

Ordinal case. A one-point distribution assigns all mass to a single category $s_j \in S$, corresponding to a CDF vector of the form

$$\mathbf{f} \in \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\} =: \{\mathbf{c}_0, \dots, \mathbf{c}_m\} \subset [0, 1]^{m+1}.$$

In each case every component satisfies $f_i \in \{0, 1\}$, so $f_i(1 - f_i) = 0$ for all i , and hence $\text{IOV} = 0$.

Conversely, $\text{IOV} = 0$ implies $\sum_{i=0}^m f_i(1 - f_i) = 0$. Since each term is nonnegative, $f_i(1 - f_i) = 0$ for all i , which forces $f_i \in \{0, 1\}$ for all i . Hence $\text{IOV} = 0$ holds *exactly* for one-point distributions.

Nominal case. A one-point distribution corresponds to a PMF vector of the form

$$\mathbf{p} \in \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\} =: \{\mathbf{e}_0, \dots, \mathbf{e}_m\} \subset [0, 1]^{m+1}.$$

In each case $\sum_{i=0}^m p_i^2 = 1$, so $\text{IQV} = 0$.

Conversely, $\text{IQV} = 0$ implies $\sum_{i=0}^m p_i^2 = 1$. Since $0 \leq p_i \leq 1$, we have $p_i^2 \leq p_i$ with equality if and only if $p_i \in \{0, 1\}$. Combined with $\sum_{i=0}^m p_i = 1$, exactly one component equals 1 and all others equal 0, i.e., $\mathbf{p} \in \{\mathbf{e}_0, \dots, \mathbf{e}_m\}$.

Maximal Dispersion

Ordinal case. Dispersion is maximal when half the probability mass lies in the lowest category and half in the highest, with all intermediate categories having probability zero. This yields

$$\mathbf{f} = \left(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2} \right)^\top,$$

for which $f_i(1 - f_i) = \frac{1}{4}$ for all i , giving $\text{IOV} = 1$.

To confirm this is the unique maximiser, note that $g(f_i) := f_i(1 - f_i)$ is strictly concave with $g(0) = g(1) = 0$ and unique maximum at

$$\frac{\partial g}{\partial f_i} = 1 - 2f_i \stackrel{!}{=} 0 \implies f_i = \frac{1}{2}.$$

Hence $\text{IOV}(\mathbf{f}) = 1 \Leftrightarrow \mathbf{f} = \left(\frac{1}{2}, \dots, \frac{1}{2} \right)^\top$.

Nominal case. Dispersion is maximal when the probability mass is distributed uniformly across all $m + 1$ categories:

$$p_i = \frac{1}{m+1}, \quad i = 0, \dots, m.$$

Then $\sum_{i=0}^m p_i^2 = \frac{1}{m+1}$, so IQV = 1.

The uniqueness of this maximiser follows from minimising $f(\mathbf{p}) := \sum_{i=0}^m p_i^2$ subject to $\sum_{i=0}^m p_i = 1$. The Lagrangian

$$\mathcal{L}(\mathbf{p}, \lambda) := \sum_{i=0}^m p_i^2 - \lambda \left(\sum_{i=0}^m p_i - 1 \right)$$

yields first-order conditions $2p_j = \lambda$ for all j , so all components are equal. The constraint then gives $p_j = \frac{1}{m+1}$ for all $j \in \{0, \dots, m\}$.

0.3.3 Skewness

A simple measure for the skewness of an ordinal process is

$$\text{skew}(\mathbf{f}) = \frac{2}{m} \sum_{i=0}^{m-1} f_i - 1.$$

0.4 Bivariate Lag Probabilities

Temporal dependence in categorical time series can be quantified through bivariate lag- h probabilities, representing the joint probability that the process takes a specific category at time t and another category at time $t + h$.

Ordinal series (cumulative).

$$f_{ij}(h) := P(X_t \leq s_i, X_{t+h} \leq s_j), \quad i, j < m.$$

Nominal series (joint).

$$p_{ij}(h) := P(X_t = s_i, X_{t+h} = s_j), \quad i, j \leq m.$$

These quantities serve as input for dependence measures such as the ordinal and nominal Cohen's κ .

0.5 Measures of Temporal Dependence

Sample Estimation

0.6 Estimators under Complete Observation

To estimate the marginal (cumulative) probabilities, the relative frequencies are expressed through a binarisation of the original process (X_t) . For ordinal data, define the indicator vector

$$\mathbf{Z}_t := (Z_{t,0}, \dots, Z_{t,m-1})^\top, \quad Z_{t,i} := \mathbb{I}\{X_t \leq s_i\},$$

and for nominal data analogously

$$\mathbf{Y}_t := (Y_{t,0}, \dots, Y_{t,m})^\top, \quad Y_{t,i} := \mathbb{I}\{X_t = s_i\}.$$

The marginal CDF and PMF vectors are then estimated by the corresponding sample means:

$$\hat{\mathbf{f}} := \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t, \quad \hat{\mathbf{p}} := \frac{1}{n} \sum_{t=1}^n \mathbf{Y}_t.$$

0.6.1 Weak Dependence Assumptions

To derive the asymptotic distribution of $\hat{\mathbf{f}}$ and $\hat{\mathbf{p}}$, a CLT for dependent data is required. Following ?, let

$$\mathcal{F}_J^L := \sigma(X_t : J \leq t \leq L)$$

denote the σ -field generated by $\{X_t : J \leq t \leq L\}$. The α -mixing coefficients of (X_t) , given strict stationarity, are

$$\alpha_{X_t}(n) := \sup_{A \in \mathcal{F}_{-\infty}^0, B \in \mathcal{F}_n^\infty} |P(A \cap B) - P(A)P(B)|.$$

The process is called α -mixing if $\alpha(n) \rightarrow 0$ as $n \rightarrow \infty$. We additionally assume *exponentially decaying* mixing coefficients: there exist constants $C > 0$ and $\rho \in (0, 1)$ such that

$$\alpha(n) \leq C\rho^n \quad \forall n \in \mathbb{N}.$$

0.6.2 Asymptotic Distribution of $\hat{\mathbf{f}}$

The indicator mappings $g_i(X_t) := \mathbb{I}\{X_t \leq s_i\}$ and $h_i(X_t) := \mathbb{I}\{X_t = s_i\}$ are measurable functions of X_t (see ?, p. 34). Measurability preserves strict stationarity, so the binarised processes $(g_i(X_t))$ and $(h_i(X_t))$ inherit strict stationarity from (X_t) . Moreover,

$$\sigma(Z_{t,i} : J \leq t \leq L) \subset \sigma(X_t : J \leq t \leq L), \quad J \leq L,$$

so $\alpha_{Z_{t,i}}(n) \leq \alpha_{X_t}(n)$ for all i and n , and hence $\mathbf{Z}_t$ and $\mathbf{Y}_t$ are also α -mixing with exponentially decaying coefficients.

The applicable CLT is due to ?, which requires strict stationarity, α -mixing, and the summability condition

$$\sum_{n=1}^{\infty} \alpha(n)^{\lambda/(2+\lambda)} < \infty \quad \text{for some } \lambda > 0.$$

Under exponential decay this is satisfied:

$$\sum_{n=1}^{\infty} \alpha(n)^{\lambda/(2+\lambda)} \leq C^{\lambda/(2+\lambda)} \sum_{n=1}^{\infty} \rho^{n\lambda/(2+\lambda)} = C^{\lambda/(2+\lambda)} \frac{\rho^{\lambda/(2+\lambda)}}{1 - \rho^{\lambda/(2+\lambda)}} < \infty.$$

The moment condition $\mathbb{E}|Z_{t,i}|^{2+\lambda} < \infty$ is trivially satisfied since $|Z_{t,i}|^{2+\lambda} \in \{0, 1\}$.

Cramér–Wold argument. By the Cramér–Wold device, $(\mathbf{Z}_t)_t$ converges in distribution to a multivariate normal if and only if every scalar projection

$$U_t := \mathbf{a}^\top \mathbf{Z}_t, \quad \mathbf{a} \in \mathbb{R}^m \setminus \{\mathbf{0}\},$$

converges to a univariate normal. Since $U_t = (\mathbf{a}^\top \circ g)(X_t)$ is a measurable function of X_t , it inherits strict stationarity and α -mixing. Moreover, $|U_t| \leq M := \max_j \left| \sum_{i=0}^j a_i \right| < \infty$, so $\mathbb{E}|U_t|^{2+\lambda} \leq M^{2+\lambda} < \infty$. Ibragimov’s CLT is therefore applicable to U_t , yielding

$$\sqrt{n}(\bar{U}_n - \mu_U) \xrightarrow{d} \mathcal{N}(0, \sigma_{\mathbf{a}}),$$

where $\mu_U = \mathbb{E}[U_t]$ and

$$\sigma_{\mathbf{a}} = \sum_{i,j} a_i a_j \left[\text{Cov}(Z_{t,i}, Z_{t,j}) + \sum_{h=1}^{\infty} \left(\text{Cov}(Z_{t,i}, Z_{t+h,j}) + \text{Cov}(Z_{t,j}, Z_{t+h,i}) \right) \right].$$

By the Cramér–Wold device, the sample mean $\hat{\mathbf{f}} = \bar{\mathbf{Z}}_n$ satisfies

$$\sqrt{n}(\hat{\mathbf{f}} - \mathbf{f}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_Z),$$

where the (i, j) -entry of Σ_Z is

$$\sigma_{Z;i,j} = \text{Cov}(Z_{t,i}, Z_{t,j}) + \sum_{h=1}^{\infty} \left(\text{Cov}(Z_{t,i}, Z_{t+h,j}) + \text{Cov}(Z_{t,j}, Z_{t+h,i}) \right).$$

Using $\text{Cov}(Z_{t,i}, Z_{t,j}) = f_{\min\{i,j\}} - f_i f_j$ and $\text{Cov}(Z_{t,i}, Z_{t+h,j}) = f_{ij}(h) - f_i f_j$, this becomes

$$\sigma_{Z;i,j} = (f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty} (f_{ij}(h) + f_{ji}(h) - 2f_i f_j).$$

0.6.3 Asymptotic Distribution of Sample Measures

Given the asymptotic normality of $\hat{\mathbf{f}}$, the delta method can be applied to derive the asymptotic properties of sample measures that are smooth functions of $\hat{\mathbf{f}}$.

IOV. The Jacobian of $\text{IOV}(\mathbf{f})$ with respect to $\mathbf{f}$ is

$$\mathbf{j}_{\text{IOV}} = \frac{4}{m} (1 - 2f_0, \dots, 1 - 2f_{m-1}),$$

giving asymptotic variance

$$\text{Var}(\widehat{\text{IOV}}) \stackrel{\text{asy.}}{=} \mathbf{j}_{\text{IOV}} \boldsymbol{\Sigma}_Z \mathbf{j}_{\text{IOV}}^\top = \frac{16}{m^2} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) \sigma_{Z;i,j}.$$

Since IOV is nonlinear, the delta method only characterises the asymptotic distribution; a finite-sample bias of order n^{-1} is present. The Hessian is

$$\mathbf{H}_{\text{IOV}}(\mathbf{f}) = \left(\frac{\partial^2 \text{IOV}}{\partial f_i \partial f_j} \right)_{i,j} = -\frac{8}{m} \mathbf{I}_m,$$

and a second-order Taylor expansion yields

$$\text{Bias}(\widehat{\text{IOV}}) = \frac{1}{2} \text{tr}(\mathbf{H}_{\text{IOV}} \text{Cov}(\hat{\mathbf{f}})) = -\frac{4}{m} \sum_{i=0}^{m-1} \sigma_{Z;i,i}.$$

The bias is strictly negative and depends only on the marginal variances $\sigma_{Z;i,i}$.

Skewness. The Jacobian of $\text{skew}(\mathbf{f})$ is

$$\mathbf{j}_{\text{skew}} = \left(\frac{2}{m}, \dots, \frac{2}{m} \right),$$

giving

$$\text{Var}(\widehat{\text{skew}}) \stackrel{\text{asy.}}{=} \frac{4}{m^2} \sum_{i,j=0}^{m-1} \sigma_{Z;ij}, \quad \mathbb{E}[\widehat{\text{skew}}] \stackrel{\text{asy.}}{=} \text{skew}(\mathbf{f}).$$

Since skewness is linear in $\mathbf{f}$, the Hessian vanishes and the estimator is exactly unbiased for all n .

0.7 Estimation under Missingness

0.7.1 Amplitude Modulation Model

Missingness is modelled via *amplitude modulation*. A binary stochastic process $(O_t)_{t \in \mathbb{Z}}$ runs concurrently with (X_t) and modulates observations:

$$O_t \in \{0, 1\}, \quad O_t = 1 \Leftrightarrow X_t \text{ is observed.}$$

The amplitude-modulating process (O_t) may be deterministic ($O_t = o_t$, known) or stochastic with $\pi_t := P(O_t = 1)$, possibly depending on the history of (X_t) , (O_t) , or external factors.

The observable process is $(O_t \mathbf{Z}_t)$, and the *naive estimator* of $\mathbf{f}$ is

$$\hat{\mathbf{f}} := \frac{1}{n} \sum_{t=1}^n O_t \mathbf{Z}_t.$$

Its expectation, by the law of iterated expectations, satisfies

$$\mathbb{E}[\hat{\mathbf{f}}] = \frac{1}{n} \sum_{t=1}^n \mathbb{E}[\mathbf{Z}_t \mathbb{E}[O_t | \mathbf{Z}_t]].$$

Under the assumption that (O_t) and (X_t) are independent, this simplifies to

$$\mathbb{E}[\hat{\mathbf{f}}] = \left(\frac{1}{n} \sum_{t=1}^n \mathbb{E}[O_t] \right) \mathbf{f}.$$

The naive estimator is therefore biased towards zero by the factor $\bar{\pi} := \frac{1}{n} \sum_{t=1}^n \mathbb{E}[O_t]$. This motivates the *corrected estimator*

$$\hat{\mathbf{f}}^* := \frac{\hat{\mathbf{f}}}{\bar{O}} = \frac{\frac{1}{n} \sum_{t=1}^n O_t \mathbf{Z}_t}{\frac{1}{n} \sum_{t=1}^n O_t}.$$

If (O_t) is deterministic then $\bar{O} = \bar{\pi}$ and $\hat{\mathbf{f}}^*$ is exactly unbiased. If (O_t) is stochastic then $\bar{O} \neq \bar{\pi}$ in general, and $\hat{\mathbf{f}}^*$ carries a finite-sample bias studied below.

0.7.2 Asymptotic Distribution of $\hat{\mathbf{f}}^*$

Assume (O_t) and (X_t) are independent, stationary, and α -mixing with exponentially decaying coefficients. Write $\mathbb{E}[O_t] = \pi$, $\gamma_O(h) := \text{Cov}(O_h, O_0)$, and $\pi(h) := \mathbb{E}[O_h O_0] = \gamma_O(h) + \pi^2$.

Introduce the extended binary vector

$$\mathbf{Z}_t^* := O_t(1, \mathbf{Z}_t^\top)^\top = (Z_{t,-1}^*, Z_{t,0}^*, \dots, Z_{t,m-1}^*)^\top,$$

with expectation

$$\mathbb{E}[\mathbf{Z}_t^*] = \pi(1, \mathbf{f}^\top)^\top =: \boldsymbol{\mu}_{\mathbf{Z}^*}.$$

By the same Cramér–Wold argument as before, the sample mean $\bar{\mathbf{Z}}_n^* := \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t^*$ satisfies

$$\sqrt{n}(\bar{\mathbf{Z}}_n^* - \boldsymbol{\mu}_{\mathbf{Z}^*}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathbf{Z}^*}),$$

where $\boldsymbol{\Sigma}_{\mathbf{Z}^*} = \text{Var}(\mathbf{Z}_t^*) + 2 \sum_{h=1}^{\infty} \text{Cov}(\mathbf{Z}_t^*, \mathbf{Z}_{t+h}^*)$. The individual entries are:

(i) $i = j = -1$:

$$\sigma_{\mathbf{Z}^*; -1, -1} = \pi(1 - \pi) + 2 \sum_{h=1}^{\infty} \gamma_O(h).$$

(ii) $j \geq 0$, $i = -1$ (by independence of (O_t) and (X_t)):

$$\sigma_{\mathbf{Z}^*; -1, j} = f_j \sigma_{\mathbf{Z}^*; -1, -1}.$$

(iii) $i, j \geq 0$:

$$\sigma_{\mathbf{Z}^*; i, j} = f_i f_j \sigma_{\mathbf{Z}^*; -1, -1} + \pi(f_{\min\{i, j\}} - f_i f_j) + \sum_{h=1}^{\infty} \pi(h) (f_{ij}(h) + f_{ji}(h) - 2f_i f_j).$$

Delta method for $\hat{\mathbf{f}}^*$. The corrected estimator is $\hat{\mathbf{f}}^* = \mathbf{g}(\bar{\mathbf{Z}}_n^*)$ where $g_j(x_{-1}, x_0, \dots, x_{m-1}) = x_j/x_{-1}$. The Jacobian of $\mathbf{g}$ evaluated at $\boldsymbol{\mu}_{\mathbf{Z}^*}$ is

$$\mathbf{G} = \begin{pmatrix} -f_0/\pi & 1/\pi & 0 & \cdots & 0 \\ -f_1/\pi & 0 & 1/\pi & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -f_{m-1}/\pi & 0 & \cdots & 0 & 1/\pi \end{pmatrix} \in \mathbb{R}^{m \times (m+1)}.$$

By the multivariate delta method,

$$\sqrt{n}(\hat{\mathbf{f}}^* - \mathbf{f}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\hat{\mathbf{f}}^*}), \quad \boldsymbol{\Sigma}_{\hat{\mathbf{f}}^*} = \mathbf{G} \boldsymbol{\Sigma}_{\mathbf{Z}^*} \mathbf{G}^\top.$$

Carrying out the matrix multiplication, the (i, j) -entry simplifies to

$$\sigma_{\hat{\mathbf{f}}^*; i, j} = \frac{1}{\pi} (f_{\min\{i, j\}} - f_i f_j) + \frac{1}{\pi^2} \sum_{h=1}^{\infty} \pi(h) (f_{ij}(h) + f_{ji}(h) - 2f_i f_j).$$

Bias of $\hat{\mathbf{f}}^*$. The Hessian of g_j at $\boldsymbol{\mu}_{\mathbf{Z}^*}$ has only two nonzero entries:

$$H_{g_j; -1, -1} = \frac{2f_j}{\pi^2}, \quad H_{g_j; -1, j} = H_{g_j; j, -1} = -\frac{1}{\pi^2}.$$

The second-order bias is then

$$\text{Bias}(\hat{f}_j^*) = \frac{1}{2} \text{tr}(\mathbf{H}_{g_j} \boldsymbol{\Sigma}_{\mathbf{Z}^*}) = \frac{f_j}{\pi^2} \sigma_{\mathbf{Z}^*; -1, -1} - \frac{1}{\pi^2} \cdot f_j \sigma_{\mathbf{Z}^*; -1, -1} = 0,$$

using $\sigma_{\mathbf{Z}^*; -1, j} = f_j \sigma_{\mathbf{Z}^*; -1, -1}$ from entry (ii) above. Hence $\hat{\mathbf{f}}^*$ is asymptotically unbiased to second order.

Special cases. Three illuminating special cases are worth noting:

1. *Full observation* ($\pi = 1$): $\sigma_{\hat{\mathbf{f}}^*; i, j} = \sigma_{Z; i, j}$, recovering the complete-data asymptotic variance.
2. *i.i.d. missingness* ($\pi(h) = \pi^2$ for all $h \geq 1$):

$$\sigma_{\hat{\mathbf{f}}^*; i, j} = \frac{1}{\pi} (f_{\min\{i, j\}} - f_i f_j) + \sum_{h=1}^{\infty} (f_{ij}(h) + f_{ji}(h) - 2f_i f_j).$$

3. *i.i.d. process* ($f_{ij}(h) = f_i f_j$ for all $h \geq 1$):

$$\sigma_{\hat{\mathbf{f}}^*; i, j} = \frac{1}{\pi} (f_{\min\{i, j\}} - f_i f_j).$$

0.7.3 Asymptotic Properties of Marginal Measures under Missingness

The delta method can be applied to $\hat{\mathbf{f}}^*$ in the same way as to $\hat{\mathbf{f}}$, yielding the asymptotic distribution of measures such as $\widehat{\text{IOV}}^*$. The corresponding Jacobians and Hessians are identical to those computed in Section 3, with Σ_Z replaced by $\Sigma_{\hat{\mathbf{f}}^*}$.

Bibliography